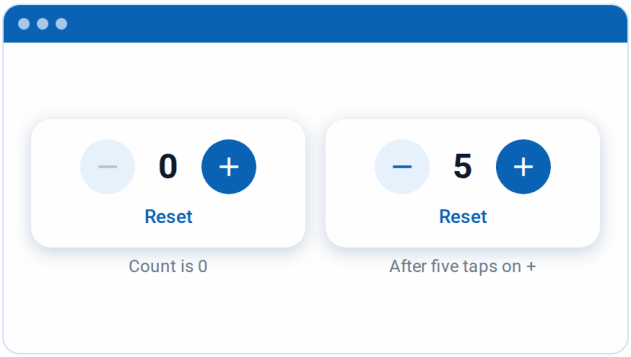
Flutter Exercises

1 Safe Counter — HARD

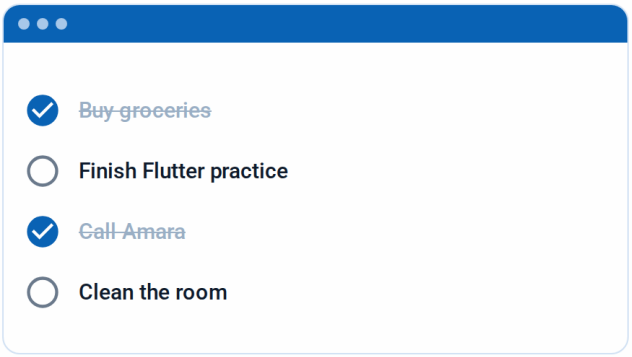
`StatefulWidget · Card · Row · IconButton · SizedBox · Text`



Recreate this and make it work: a Card with a minus button, a number, a plus button and a "Reset" link. The number can never go below 0, and when it is 0 the minus button must look and act disabled (onPressed: null). Reset sets it back to 0.

2 Todo Checklist — HARD

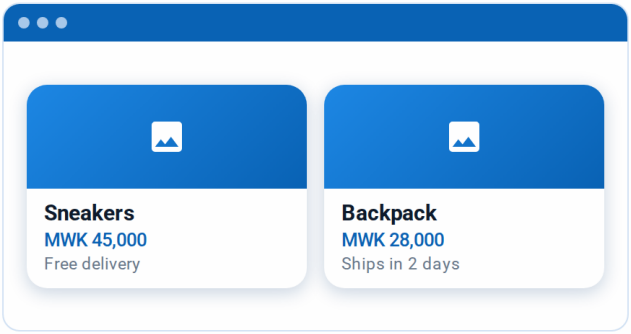
`ListView · ListTile · Icon · GestureDetector · StatefulWidget`



Recreate this and make it work: a ListView of four tasks. Tapping a task toggles it between done (filled blue check, struck-through grey text) and not done (grey empty circle, normal text). Keep the done/not-done state in a list of booleans.

3 Product Comparison — HARD

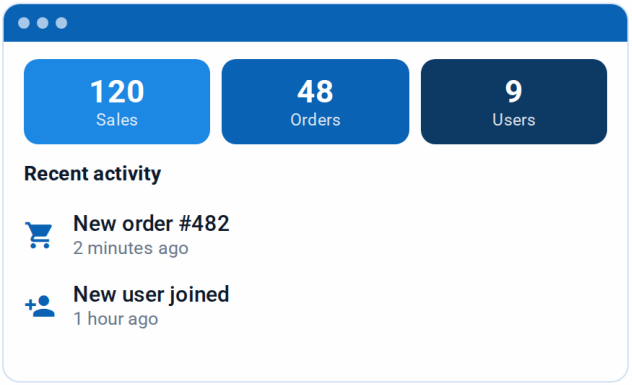
`Row · Expanded · Card · Column · Image · SizedBox · Padding`



Recreate this: two product Cards side by side that always share the width equally (Expanded), with a 12 pixel SizedBox gap between them. Each has an Image on top, a bold name, a blue price and a small grey note, and the page has 16 pixels of padding on the sides.

4 Stats Dashboard — HARD

`Column · Row · Expanded · Container · ListView · ListTile`

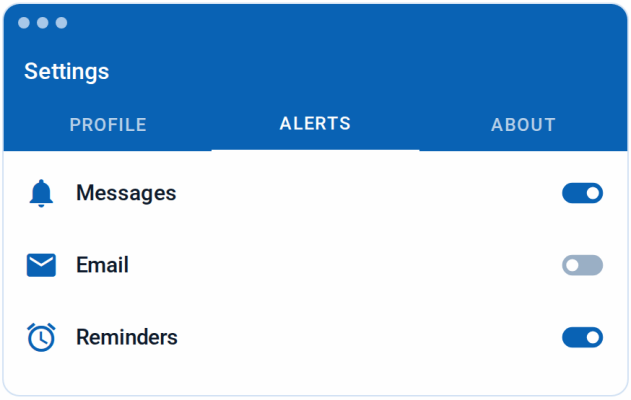


Recreate this: three equal stat boxes in a Row (Expanded), a bold "Recent activity" heading below them, and a ListView with two ListTiles (icon, title, time) filling the remaining space.

5 Tabbed Settings

HARD

`Scaffold · AppBar · TabBar · TabBarView · ListView · ListTile`

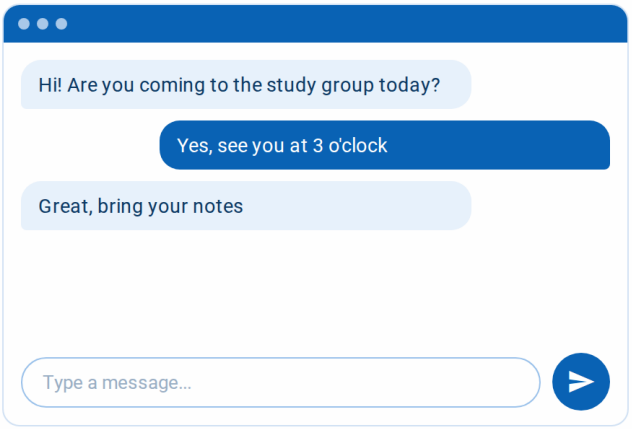


Recreate this: an AppBar with a TabBar of three tabs (Profile, Alerts, About). The "Alerts" tab opens a ListView of three ListTiles, each with a leading icon, a title and a trailing on or off icon that flips when the tile is tapped. The other two tabs can hold centered text.

6 Chat Bubbles

HARD

`ListView · Row · Expanded · Container · Padding · TextField · IconButton`

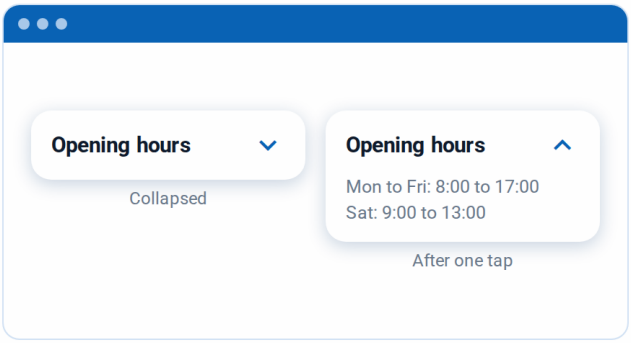


Recreate this chat screen: received messages are light and aligned left, sent messages are blue and aligned right, and neither bubble is ever wider than three quarters of the screen (Expanded with flex 3 next to an empty Expanded with flex 1). A TextField and a send IconButton stay at the bottom.

7 Expandable Card

HARD

`Card · GestureDetector · Column · Icon · StatefulWidget`

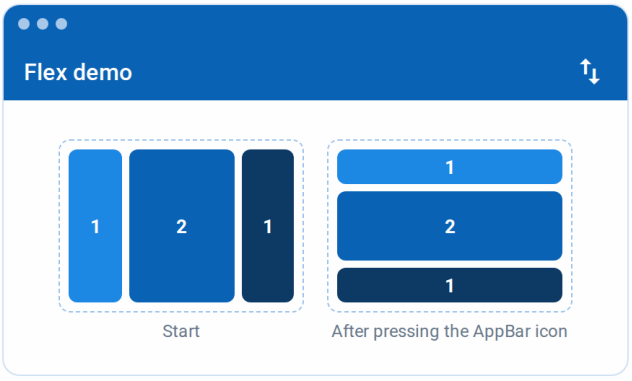


Recreate this and make it work: a Card whose header row (title plus arrow icon) is wrapped in a GestureDetector. Each tap shows or hides the details underneath and flips the arrow between expand_more and expand_less.

8 Direction Switcher

HARD

`Flex · Expanded · AppBar · IconButton · StatefulWidget`

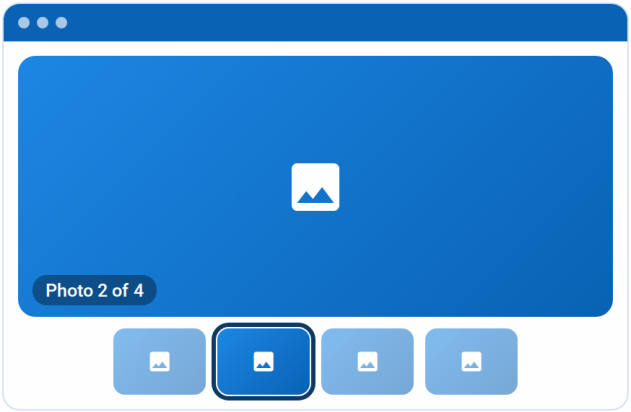


Recreate this and make it work: three Expanded boxes with flex 1, 2 and 1 inside a Flex widget. The IconButton in the AppBar flips the Flex direction between Axis.horizontal and Axis.vertical while the proportions stay the same.

9 Photo Viewer

HARD

`Column · Expanded · Image · ListView · GestureDetector · StatefulWidget`

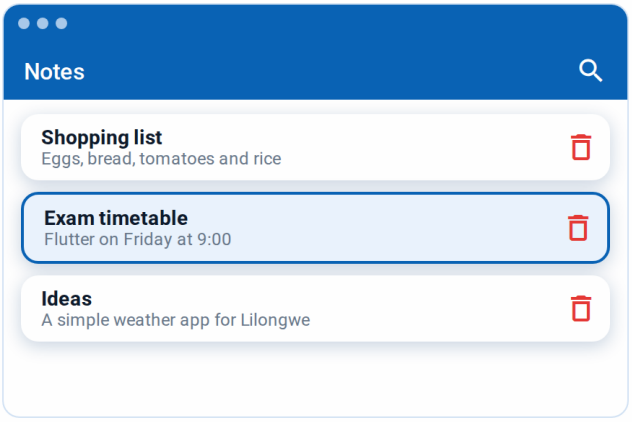


Recreate this and make it work: a large Image that fills the available height (Expanded), with a horizontal ListView of four thumbnails underneath. Tapping a thumbnail makes it the big image, gives that thumbnail a dark outline, and updates the "Photo 2 of 4" label.

10 Notes App (capstone)

HARD

`Scaffold · AppBar · ListView · Card · ListTile · GestureDetector · IconButton · Center`

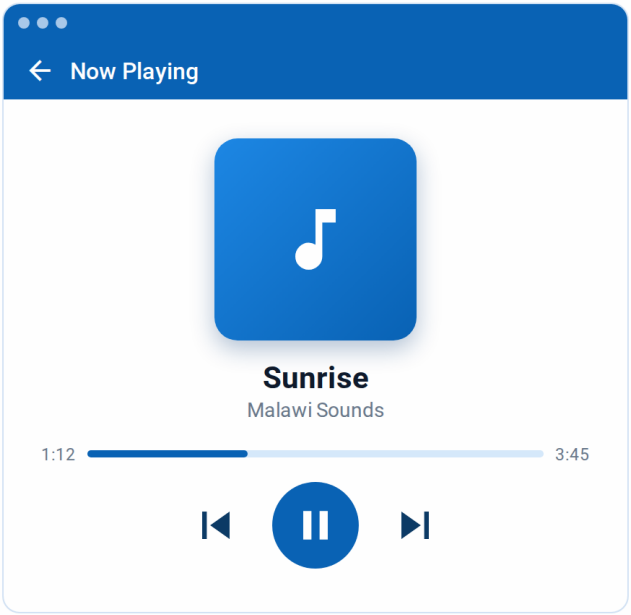


Recreate this and make it work: a ListView of note Cards. Tapping a card highlights it with a blue border (only one at a time, using GestureDetector), and the red delete IconButton removes that note. When the list is empty, show a centered "No notes yet" text instead.

11 Music Player (capstone)

HARD

`Scaffold · AppBar · SafeArea · Column · Image · Row · Expanded · IconButton · SizedBox · Container`

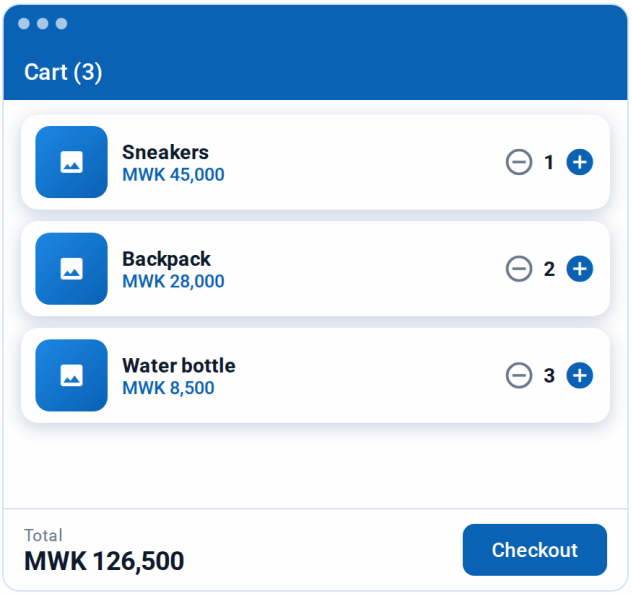


Recreate this player screen: an AppBar with a back arrow, a square album Image, a bold title and grey artist, a progress bar built from two nested Containers with the time labels on each side (Expanded), and three control IconButtons. The big button toggles between play and pause when pressed.

12 Shopping Cart (capstone)

HARD

`Scaffold · AppBar · ListView · Card · Row · Expanded · IconButton · Container · Text`



Recreate this cart screen and make it work: a ListView of Cards, each with an Image, a name and price, and a quantity stepper made of two IconButtons. A bar fixed at the bottom shows the total, which must update every time a quantity changes. Quantity cannot go below 1.